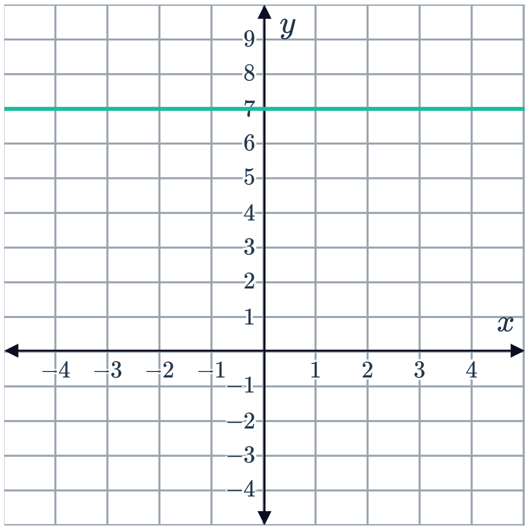
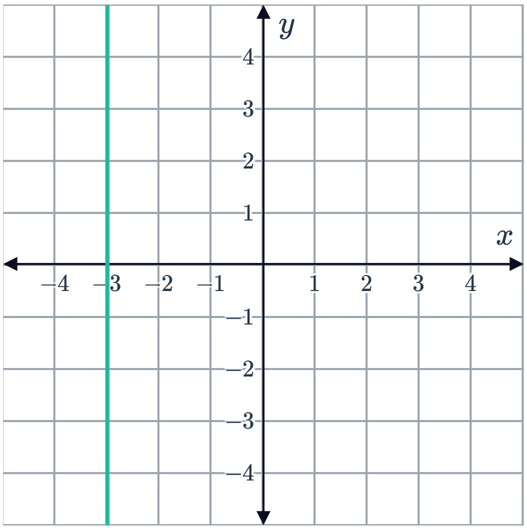
Gradient of a line

1 Describe the gradient of the lines in the following graphs:

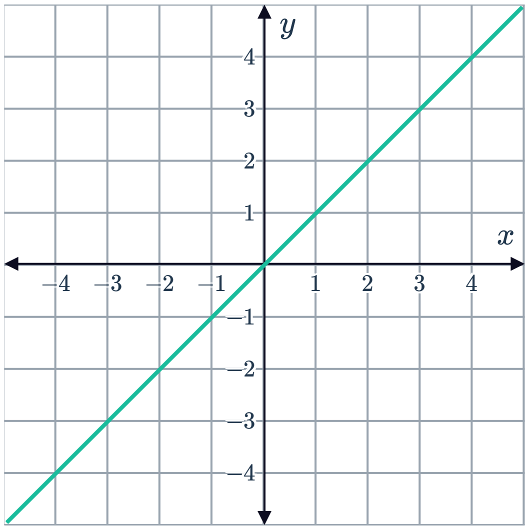
a



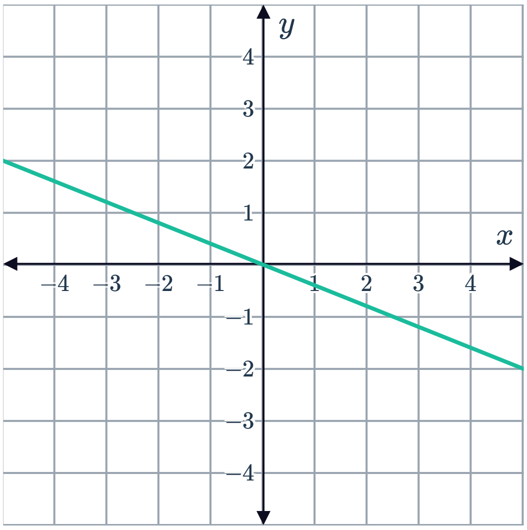
b



c



d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

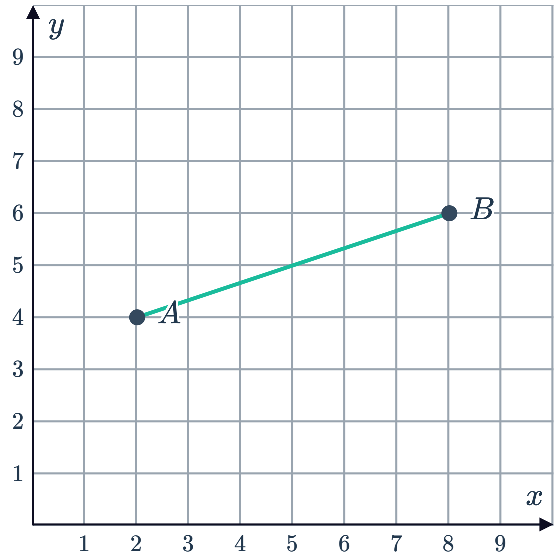
2 For each of the following intervals betwee[redacted]nts A and B:

i Find the rise.

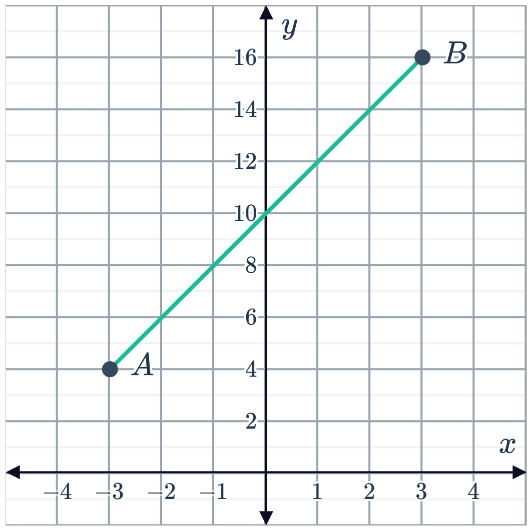
ii Find the run.

iii Find the gradient.

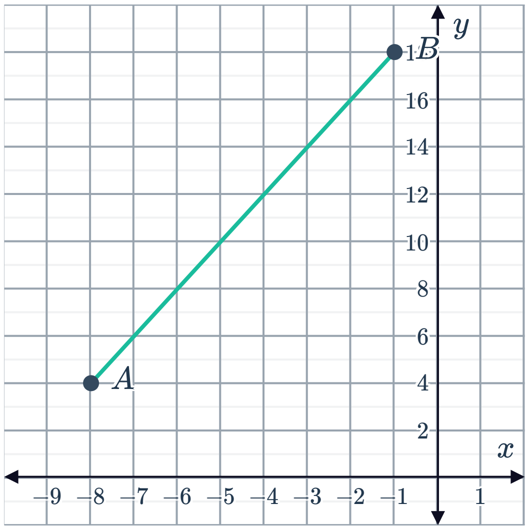
a $A\,(2,4)$ and $B\,(8,6)$



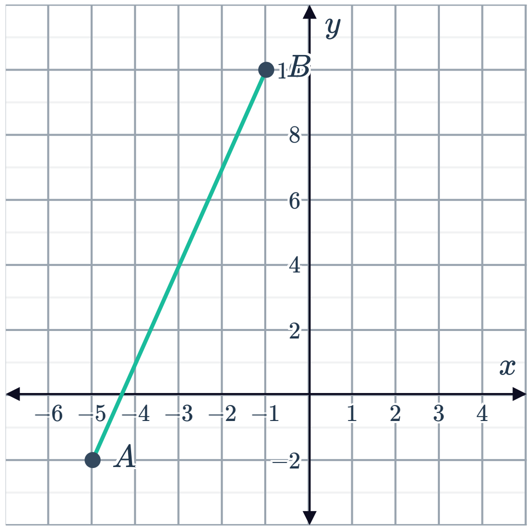
b $A\,(-3,4)$ and $B\,(3,16)$



c $A\,(-8,4)$ and $B\,(-1,18)$

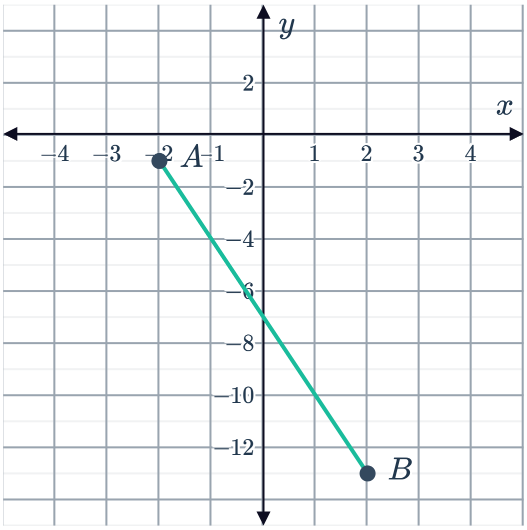


d $A\,(-5,-2)$ and $B\,(-1,10)$

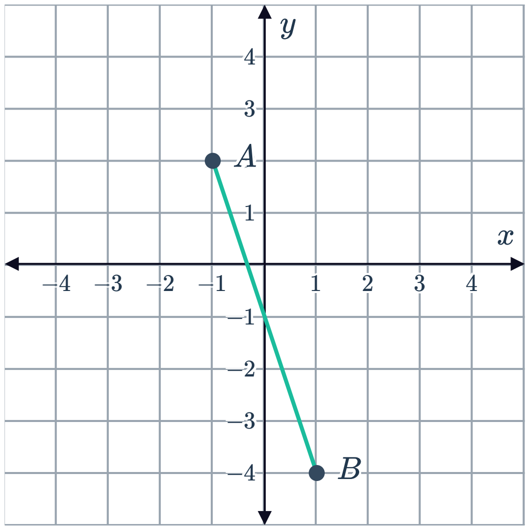


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

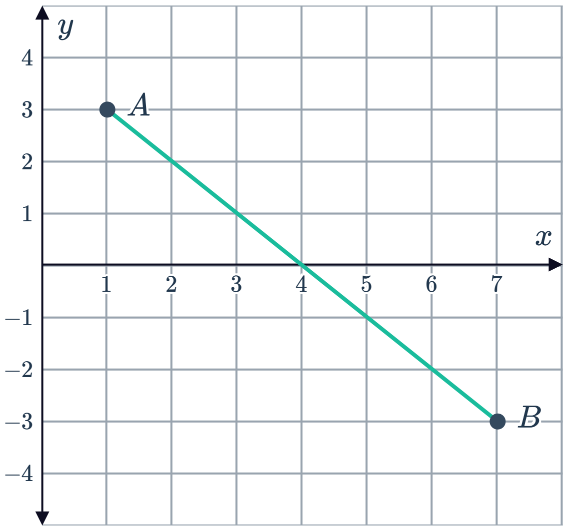
e $A(-2,-1)$ and $B(2,-13)$



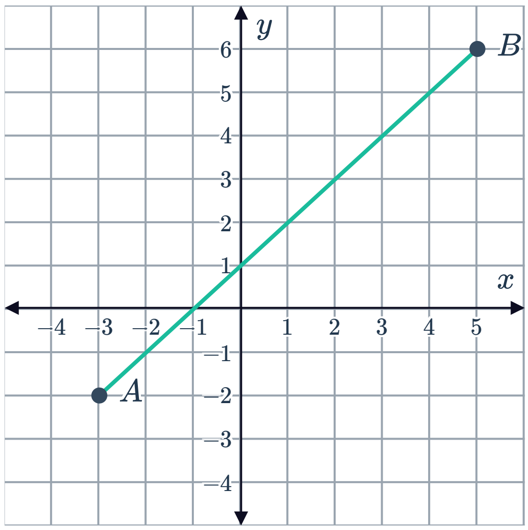
f $A(-1,2)$ and $B(1,-4)$



g $A(1,3)$ and $B(7,-3)$



h $A(-3,-2)$ and $B(5,6)$

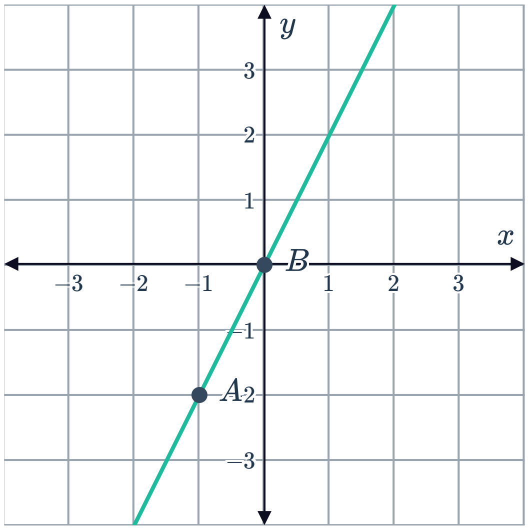


3 Explain why a vertical line has an undefined gradient.

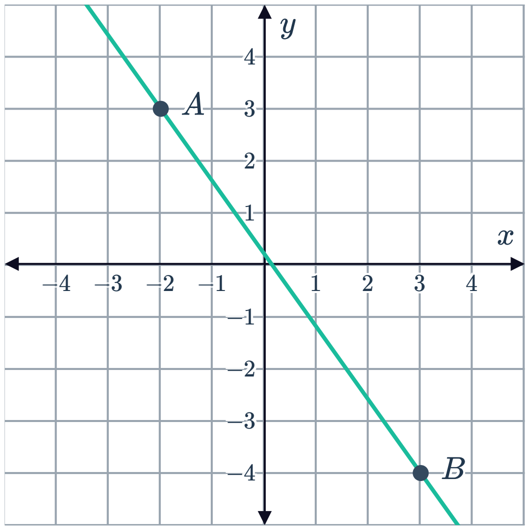
4 State the gradient of any line parallel to the x-axis.

5 Find the gradient of the following lines pa[illegible] through points A and B:

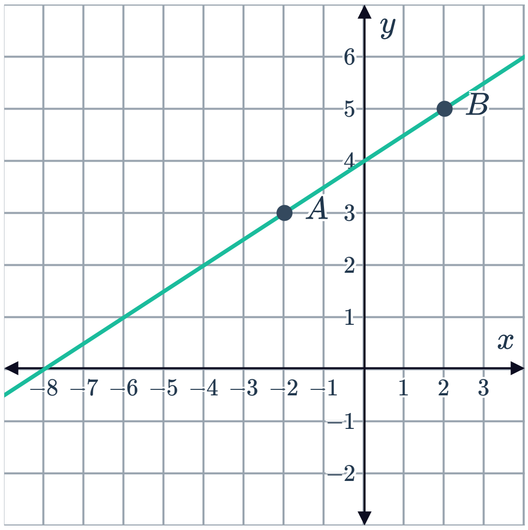
a



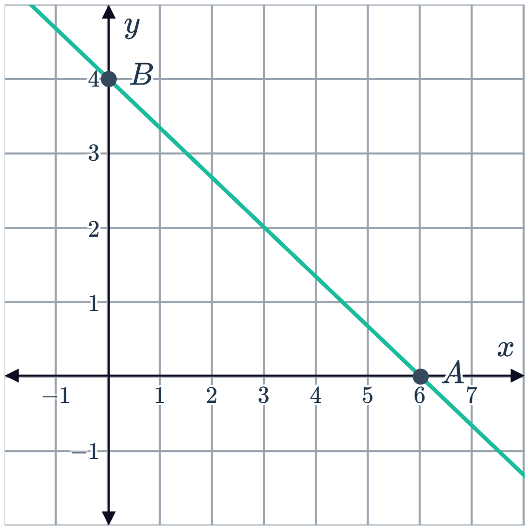
b



c



d



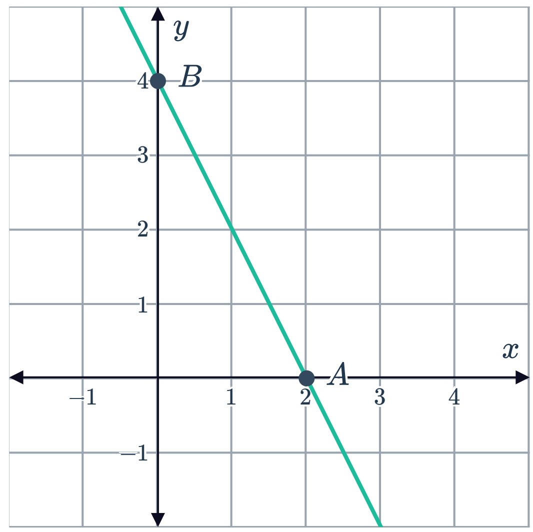
Gradient formula (Extended)

6 Find the gradient of the interval joining $A\,(-9, 4)$ and $B\,(-3, -5)$.

7 If we have two points and the slope formula $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2)?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Consider the line plotted, where $A\,(2, 0)$ and $B\,(0, 4)$ both lie on the line.



a Solve for the gradient of the line.

b As x increases, what happens to the value of y?

9 Find the gradient of the line that passes through the given points:

a $(-1, 0)$ and $(0, 3)$

b $(-4, 7)$ and $(1, 10)$

c $(1, -4)$ and the origin

d $(2, -6)$ and the origin

e $(6, 4)$ and $(3, 4)$

f $(-6, 5)$ and $(4, 5)$

g $(-2, -5)$ and $(-9, -12)$

h $(-3, -1)$ and $(-5, 1)$

10 Consider the points $A\,(-11, -9)\,, B\,(-5, 1)$ and $C\,(-2, 6)$:

a Find the gradient of AB.

b Find the gradient of BC.

c Do the points A, B and C lie in a straight line?

11 Consider the following points: $A\,(26, m - 24)$, $B\,(-1, m)$ and $C\,(-10, 9)$.

Find m, given that A, B and C are collinear.

12 Given the gradient of the line passing through the two points, find the value of the pronumeral:

a $(4, -3)$ and $(1, t)$, gradient $= -2$

b $(5, 3)$ and $(2, t)$, gradient $= -4$

c $(5, 3)$ and $(d, 63)$, gradient $= 4$

d $(11, c)$ and $(-20, 16)$, gradient $= -\dfrac{4}{7}$

13 The line $x = 4$ intersects the line $y = 2x - 10$ at the point Q.

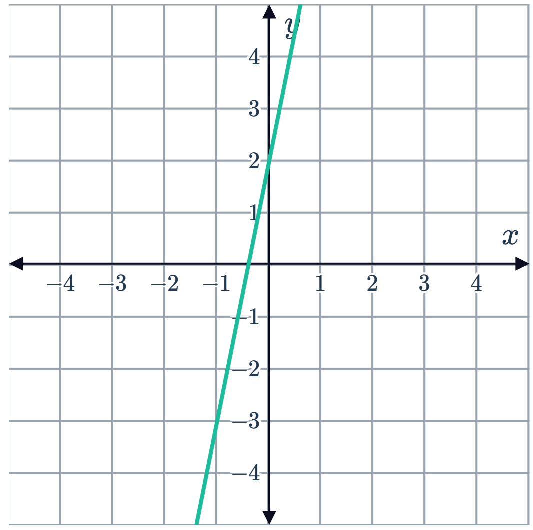
The line $x = -4$ intersects the line $y = 2x - 10$ at point R.

a Find the coordinates of Q.

b Find the coordinates of R.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 Consider the line $y = 5x + 2$ graphed:



a Find the y-value of the point on the line where $x = 5$.

b Find the gradient of the line.

15 Consider the line $y = -2x + 8$.

a Find:

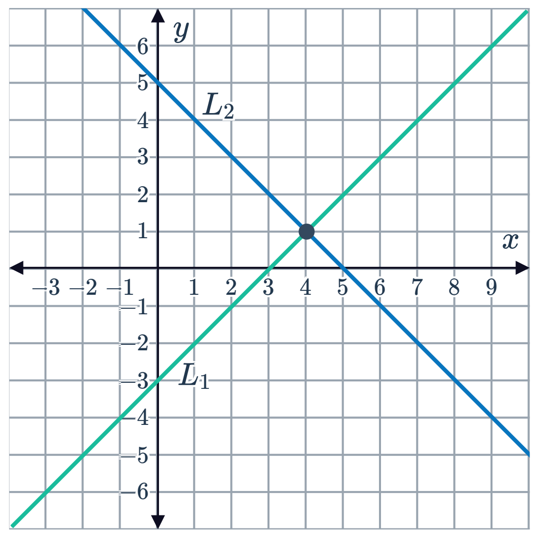
i y-intercept

ii x-intercept

b Sketch the line on a number plane.

c Hence, find the gradient of the line.

16 Two lines L_1 and L_2 have equations $y = x - 3$ and $y = -x + 5$ respectively. The lines and their point of intersection have been graphed:



a When $x = 4$, find the y-coordinate of the corresponding point on L_1.

b When $x = 4$, find the y-coordinate of the corresponding point on L_2.

c Find the gradient of the two lines:

i L_1

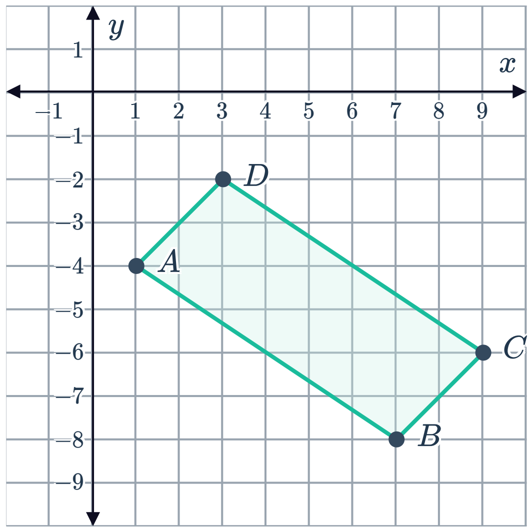
ii L_2

d Find the product of their gradients.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Properties of polygons (Extended)

17 Consider the quadrilateral $ABCD$ that has been graphed on the number plane:



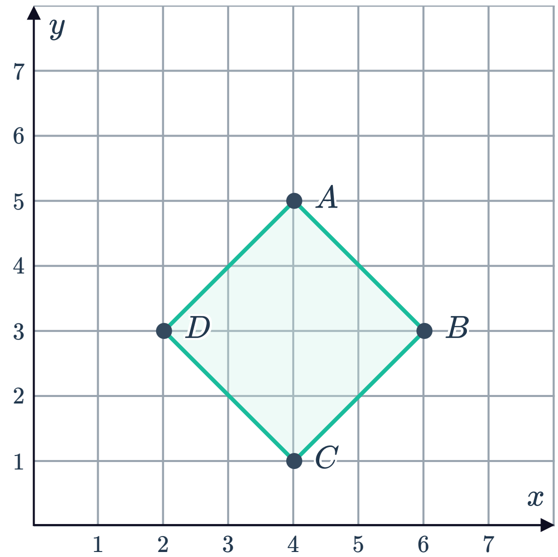
a Find the gradient of following sides:

i AB ii CD

iii AD iv BC

b What type of quadrilateral is $ABCD$? Explain your answer.

18 The 4 vertices of square $ABCD$ have been plotted on a number plane.



a Find the gradient of side AB.

b Find the gradient of side BC.

c Find the product of the gradients in parts (a) and (b).

d If two lines are perpendicular their gradients multiply to -1. Are sides AB and BC perpendicular?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 The points $P\,(0,-2)$, $Q\,(-2,0)$, $R\,(0,4)$ an[redacted]$,2)$ are graphed below:

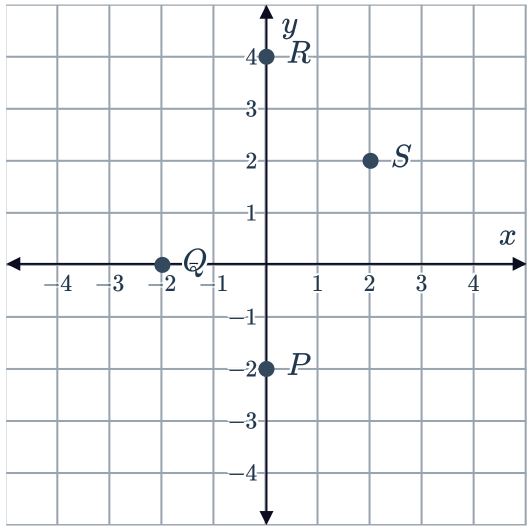
a Find the gradient of PQ.

b Find the gradient of RS.

c Are PQ and RS parallel?

d Are QR and PS parallel?

e Identify the type of quadrilateral $PQRS$ is.



20 The vertices of $\triangle ABC$ are $A\,(9,-12)$, $B\,(4,4)$ and $C\,(-8,-5)$. The sides AB and AC have midpoints D and E respectively.

a Find the coordinates of points D and E.

b Find the gradient of side BC.

c Find the gradient of side DE.

d Are BC and DE parallel to each other?

21 $A\,(-4,-2)$, $B\,(2,1)$ and $C\,(2,-4)$ are the vertices of a triangle.

a Name the side of the triangle that is a vertical line.

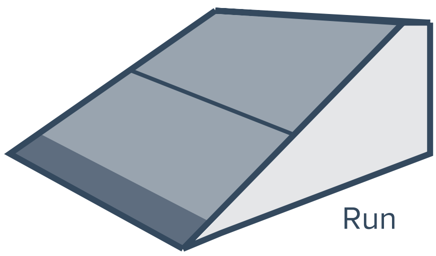
b Find the area of the triangle.

Applications

22 Consider the following ramp:

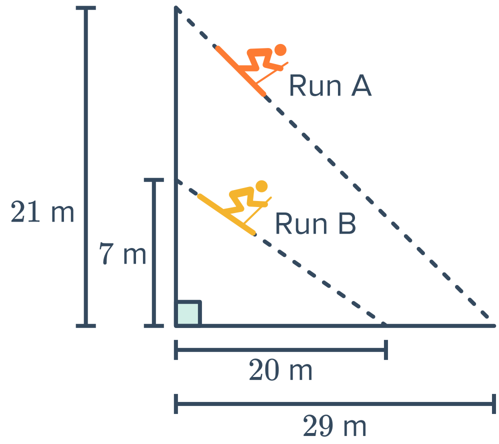
a Find the gradient of this skateboard ramp if it rises 0.9 metres above the ground and runs 1 metre horizontally at the base.

b The ramp can only be used as a 'beginner's ramp' if for every 1 metre horizontal run, it has a rise of at most 0.5 metres. Can it be used as a 'beginner's ramp'?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23 A certain ski resort has two ski runs as shown in the diagram:



- **a** Find the gradient of Run A. Round your answer to two decimal places.
- **b** Find the gradient of ski run B. Round your answer to two decimal places.
- **c** Which run is steeper?

24 A paratrooper falls to the ground along a diagonal line. His fall begins $1157\ \mathrm{m}$ above the ground, and the line he follows has a gradient of 1.3. That is, he falls $1.3\ \mathrm{m}$ vertically for every $1\ \mathrm{m}$ he moves across horizontally.

How far horizontally across the ground does he land from his initial position in the sky?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647